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Practical work of Image Processing

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Japanese French Week on Mechatronics, Sep. 14th – 22nd, 2026

Practical work using Matlab



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Physics, Informatics and Robotics

(A) Find the location of an object in a picture.

- 1) Take a picture of some object using your camera.
- 2) Send the picture to Matlab.
- 3) Set a threshold to separate the object from background.
- 4) Binarize the image.
- 5) Execute the projection to x - and y - direction.
- 6) Calculate the center coordinate of the object.

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(B) Compare the effect of the averaging filter with the median filter.

- 1) Take a picture of some texture or with spiking noise.
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- 3) Find the averaging/uniform filter and the median filter.
- 4) Apply the filters to the picture.
- 5) Examine the difference between the two filtered images.

(C) Examine the effects of different edge-extraction filters.

- 1) Take a picture of some objects having textures.
- 2) Send the picture to Matlab.
- 3) Find different edge-detection filters.
- 4) Apply the filters to the picture.
- 5) Examine the difference among the filtered images.

1. Linear Algebra
2. Signals and Systems
3. Digital Signal Processing
4. Probability Theory and Random Process
5. C/C++, Python Programming Skill, Matlab

Computational Intelligence

Machine Learning

Data Science

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Digital Image Formation

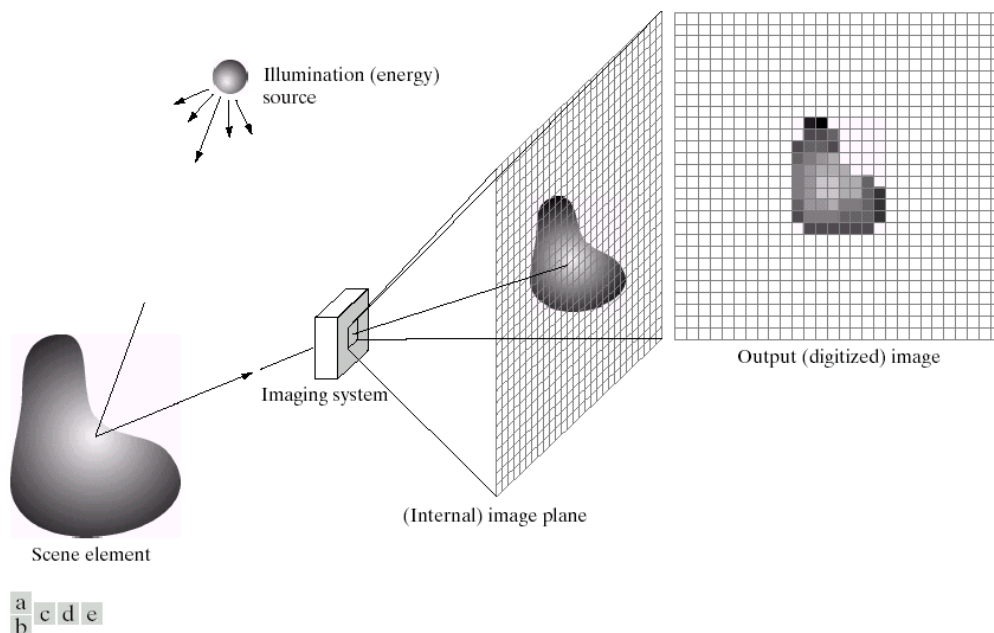


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy (“illumination”) source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

A **digital image** is a representation of a two-dimensional image as a finite set of digital values, called picture elements or pixels.

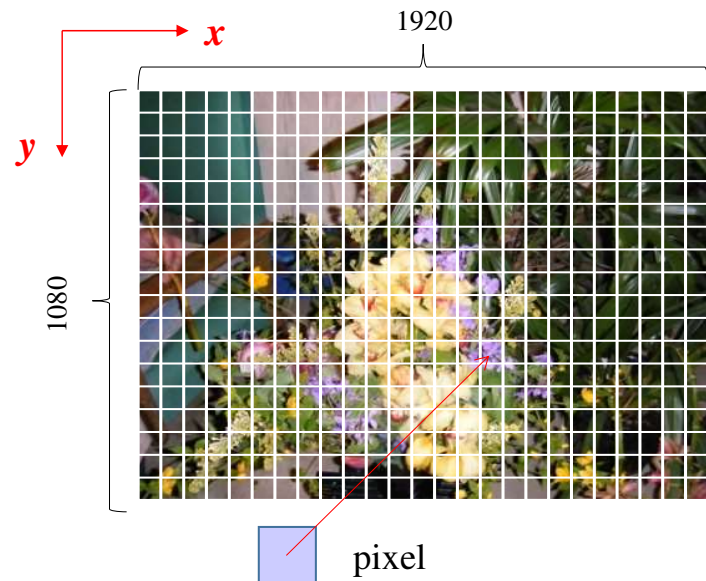
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Description of Image Data

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Image Size: 1920×1080 , 1280×720 ,
 640×480 , 320×240

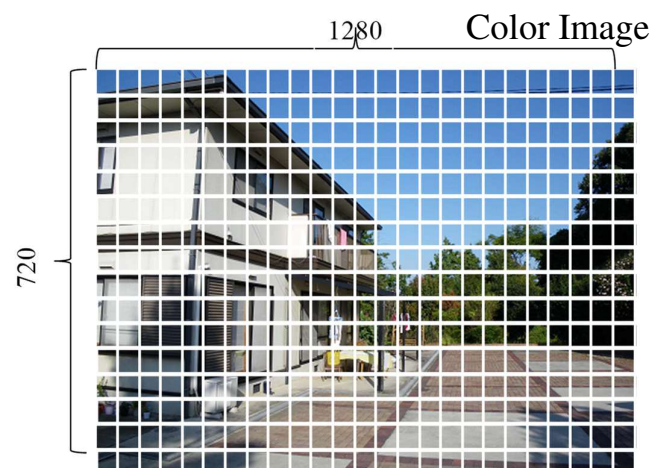
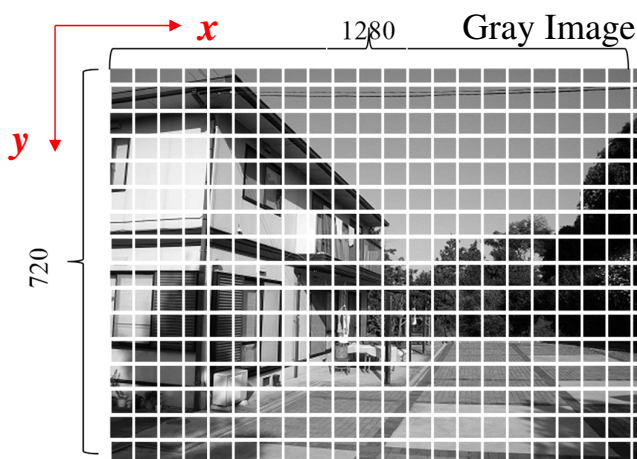


When you take a picture using a digital camera or a mobile phone, the image size is described such as 1280×720 and 1920×1080 . It means that the image taken by the camera consists of the 1920×1080 picture elements. A picture is divided into 1920×1080 small pixels.

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Description of Image Data

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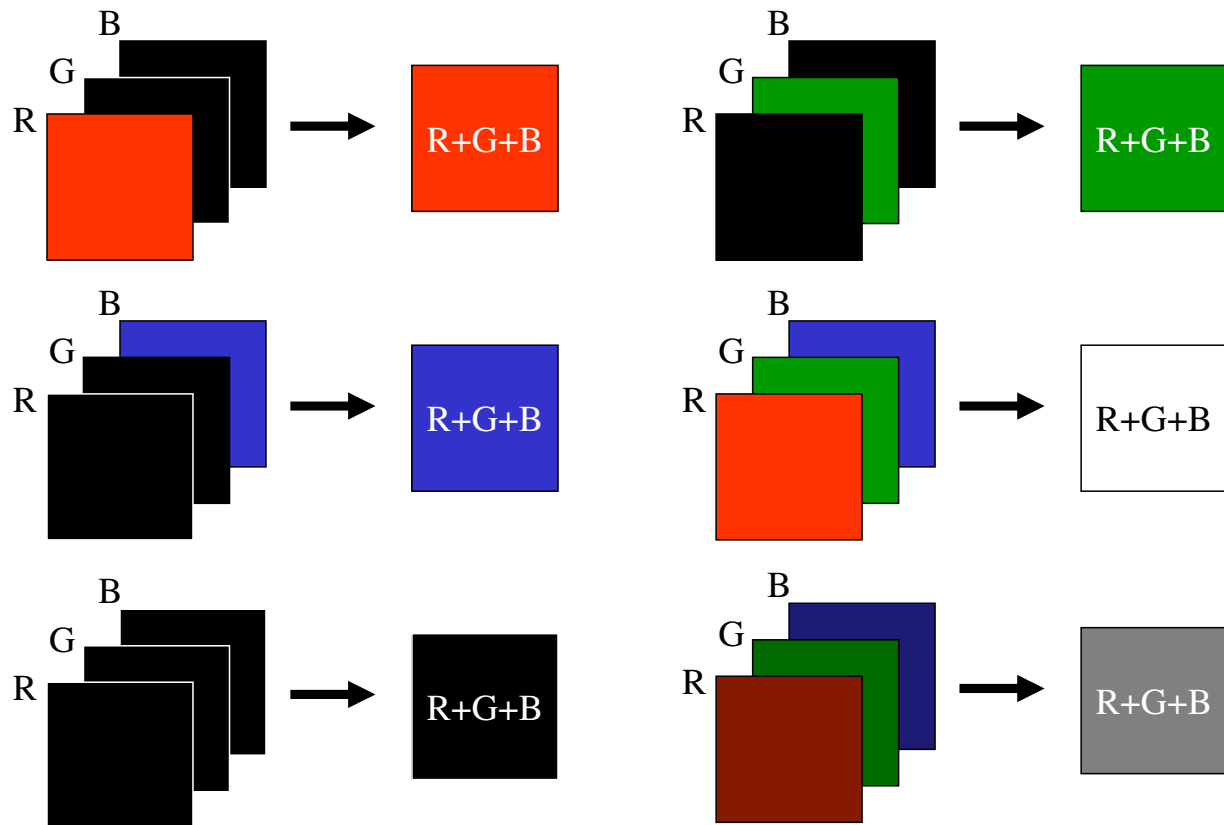
Each pixel of a gray-scale image or a monochrome image has a brightness value. In a computer, the brightness value is expressed by an integer from 0 to 255, where 0 is black and 255 is white.

In a color image, each pixel consists of 3 data. There are different ways to express color data, however RGB expression is widely used. R means red color, G means green color and B means blue color, which are the three primary colors of light. The three values of RGB colors are expressed also by the number from 0 to 255. Any color that can be displayed in a color display is expressed by the combination of RGB data.

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Expression of Color Image Data

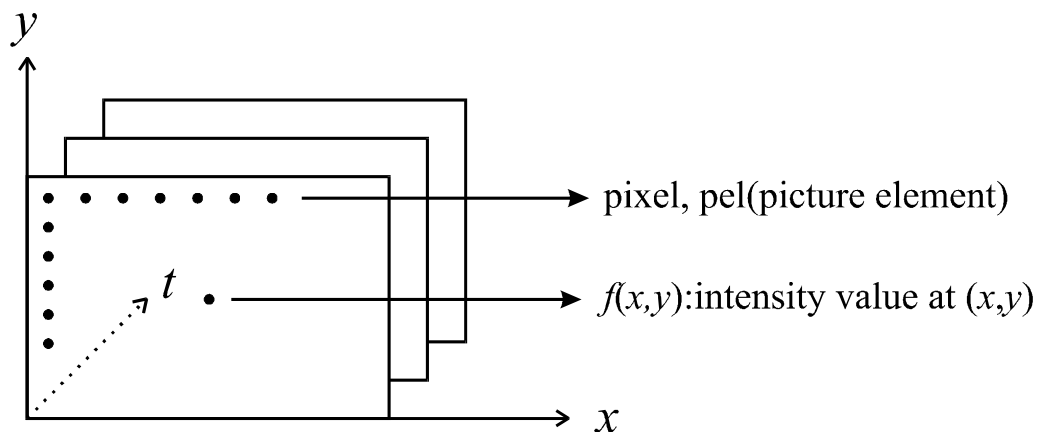
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Description of Image Data

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picture size $\Rightarrow$ picture resolution ; 640×480, 1980×1080

$0 \leq f(x, y) \leq L (= 255)$; gray level, 8bit/pixel

(x, y) ; spatial coordinate

t ; temporal coordinate

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Algebraic Description

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$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0, N-1) \\ f(1,0) & f(1,1) & \cdots & f(1, N-1) \\ \vdots & & & \\ f(M-1,0) & \cdots & \cdots & f(M-1, N-1) \end{bmatrix} \quad M \times N \text{ matrix}$$

$$\mathbf{A} = \begin{bmatrix} a_{00} & a_{01} & \cdots & a_{0,N-1} \\ \vdots & & & \\ \vdots & & & \\ a_{M-1,0} & \cdots & \cdots & a_{M-1,N-1} \end{bmatrix} \quad a_{ij} = f(x=i, y=j)$$

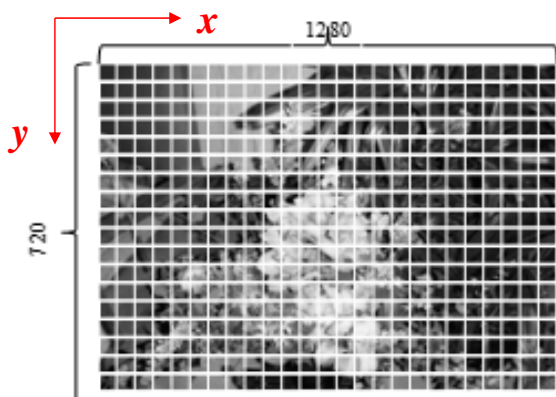
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Image Description in C-program

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Image data are described as 2 dimensional array, which elements have values from 0 to 255. As a gray image, typical expression is as follows;

```
unsigned char image-name[y-pixel-size][x-pixel-size];
```



Left image presents 1280x720 pixels picture in gray scale values, and the description of the image is;

```
unsigned char flower[720][1280];
```

or

```
unsigned char image_name[HEIGHT][WIDTH];
```

where HEIGHT is 720 and WIDTH is 1280.

For color images, each pixel consists of three colors, which are **R**, **G** and **B** that represent **Red**, **Green** and **Blue**, respectively.

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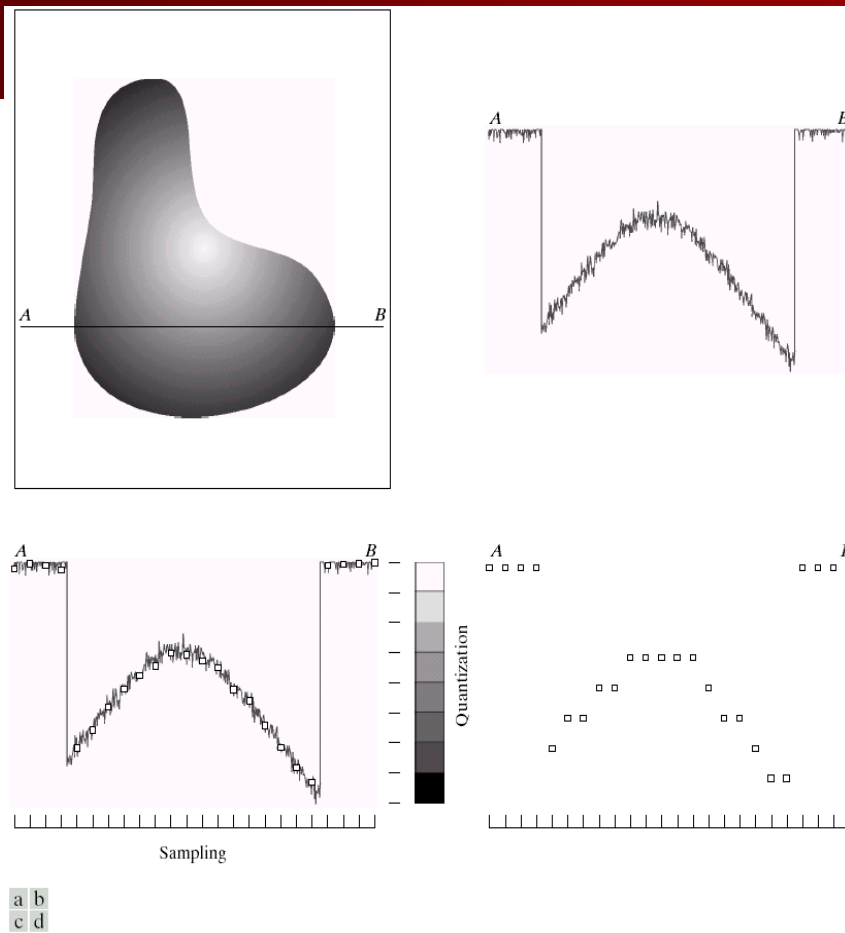


FIGURE 2.16 Generating a digital image. (a) Continuous image. (b) A scan line from *A* to *B* in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

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Image Quantization

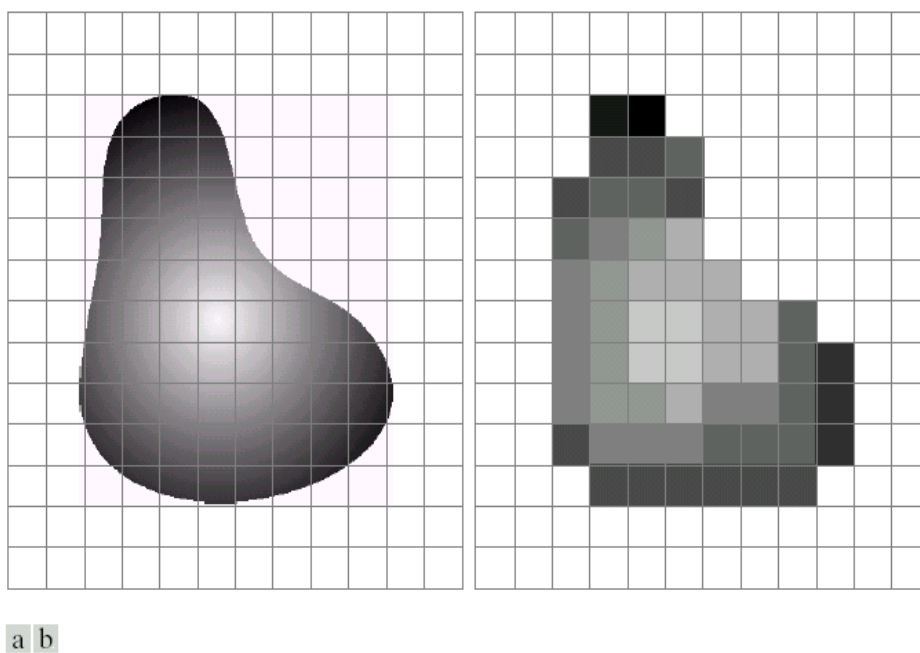
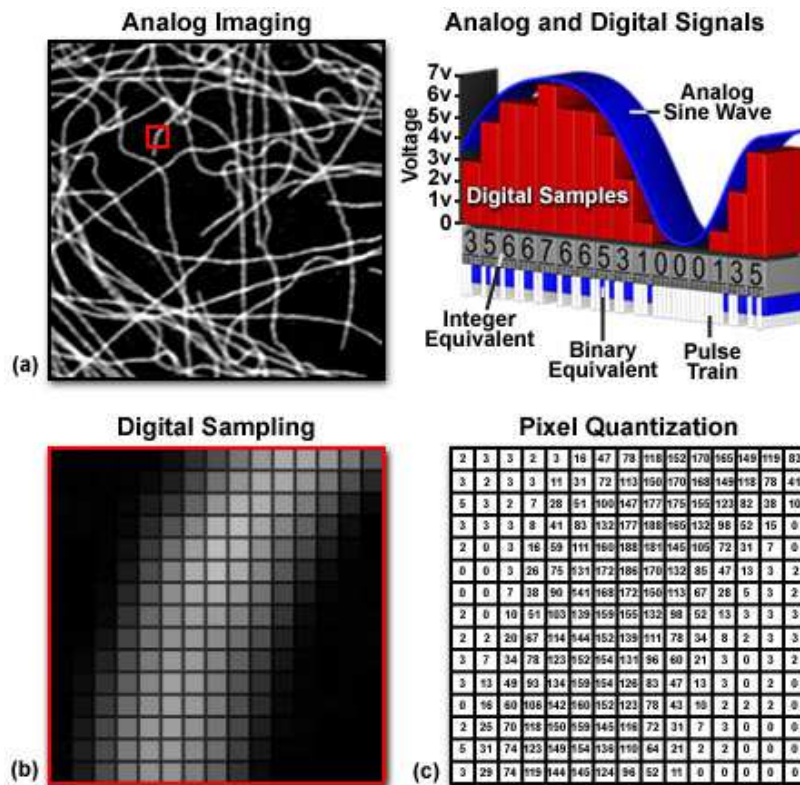


FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

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Image Quantization

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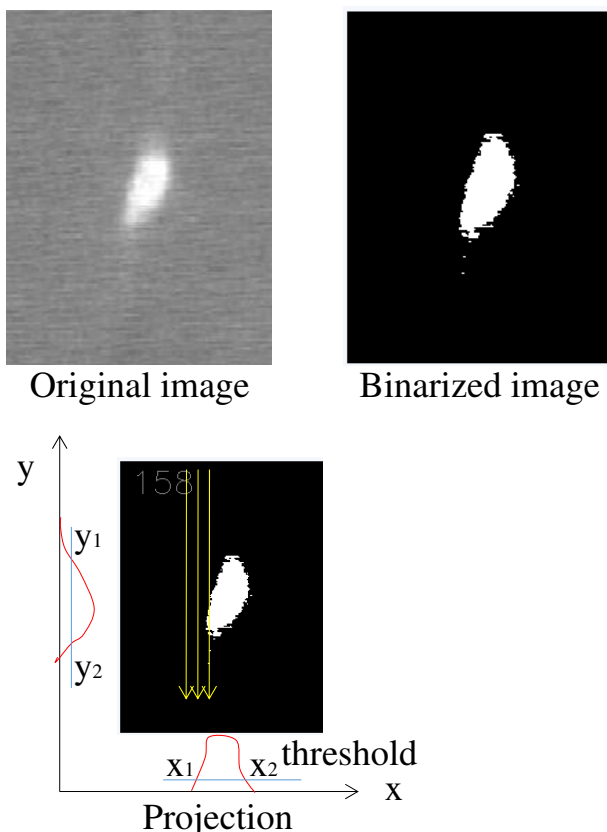


<http://learn.hamamatsu.com/articles/microscopyimaging.html>

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Projection for the Localization of an Object

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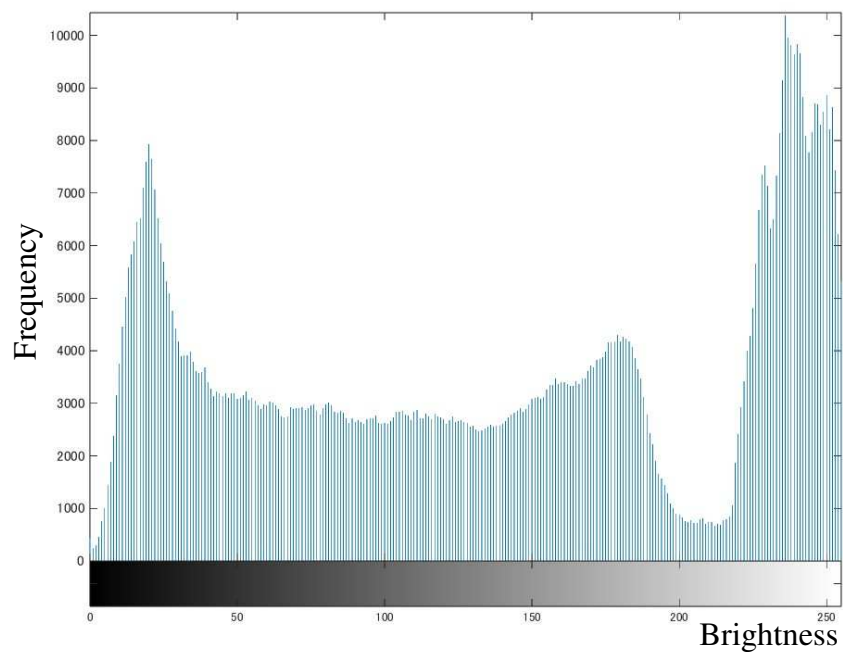
“Projection” is the operation to count white pixels along x-axis and y-axis. The counting result shows the distribution of white pixels. The area that exceeds a certain threshold should be the position, or the region of the white blob.

We can say that a white blob exists in the area between (x_1 and x_2) and (y_1 and y_2).

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Histogram

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```
I = imread("Temple.jpg");
imshow(I);
```

```
imhist(I);
```

```
imwrite(I, "Histogram.png")
```

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Adjusting Image Contrast

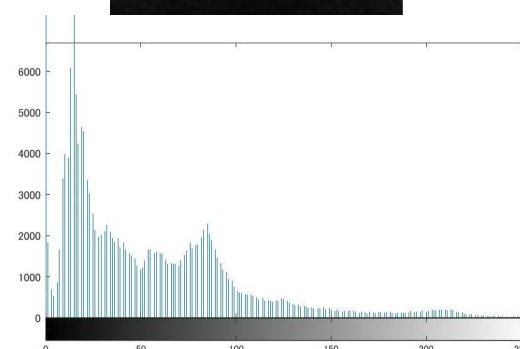
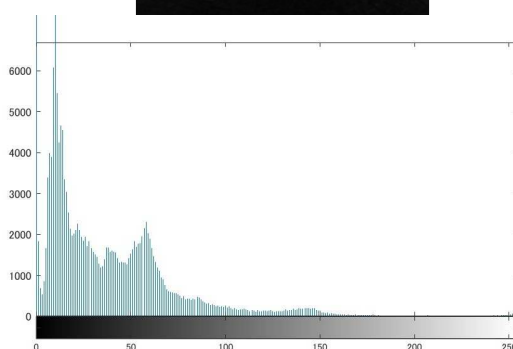
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```
imadjust(gs);
```



Non-linear
adjustment



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Image Filtering by Convolutional Operations

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Averaging/Smoothing/Uniform filter

$$h = \begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$H = \text{fspecial}(\text{"average"}, 3)$

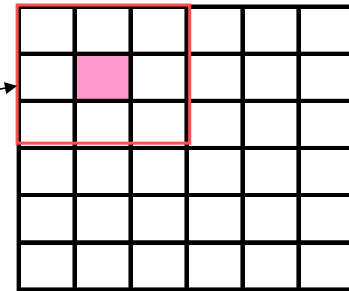


Image data

$W_{-1,-1}$	$W_{0,-1}$	$W_{+1,-1}$
$W_{-1,0}$	$W_{0,0}$	$W_{+1,0}$
$W_{-1,+1}$	$W_{0,+1}$	$W_{+1,+1}$

$W_{mn} =$

Kernel

$$F_{ij} = \sum_{m=-k}^{+k} \sum_{n=-k}^{+k} W_{mn} * f_{i+m,j+n}$$

Operation

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Geometric translation

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$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

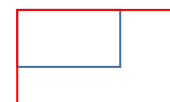
Shift in x- or y-direction

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{pmatrix}$$



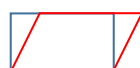
Scale translation

$$\begin{pmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



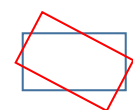
Shear deformation

$$\begin{pmatrix} 1 & \tan \theta_y & 0 \\ \tan \theta_x & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



Rotation

$$\begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

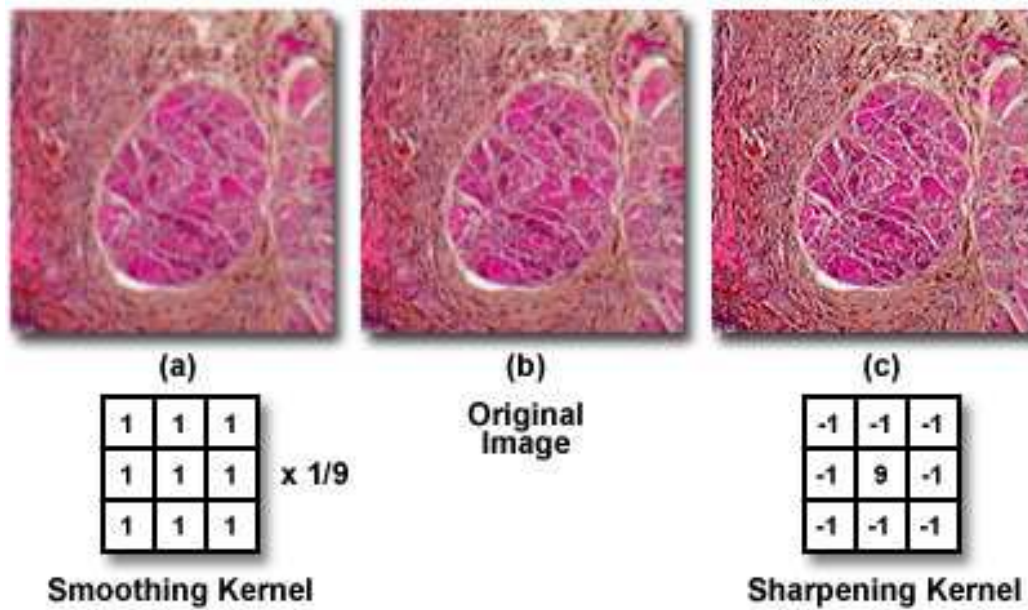


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Image Filtering by Convolutional Operations

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Smoothing and Sharpening Convolution Kernels



microscopy.fsu.edu

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Edge Detection

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Sobel operator

$$h_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$h_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$h_{xy} = (h_x^2 + h_y^2)^{\frac{1}{2}}$$

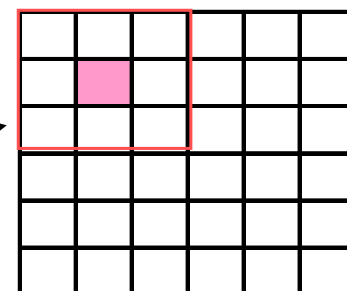


Image data

H = fspecial("sobel")

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Edge Detection

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$$h_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad h_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

10	10	10	50	50	50
10	10	10	50	50	50
10	10	10	50	50	50
10	10	10	50	50	50
10	10	10	50	50	50
10	10	10	50	50	50

$$h_x = \begin{aligned} & -1 * 10 + 0 * 10 + 1 * 10 \\ & + (-2 * 10) + 0 * 10 + 2 * 10 \\ & -1 * 10 + 0 * 10 + 1 * 10 \end{aligned}$$

$$h_x = \begin{aligned} & -1 * 10 + 0 * 10 + 1 * 50 \\ & + (-2 * 10) + 0 * 10 + 2 * 50 \\ & -1 * 10 + 0 * 10 + 1 * 50 \end{aligned}$$

$$h_{xy} = (h_x^2 + h_y^2)^{\frac{1}{2}}$$

0	160	160	0
0	160	160	0
0	160	160	0
0	160	160	0

EdgeDetect = imfilter(GrayImage, H);

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Edge Detection

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Laplacian filter

$$h_x = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image data

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Edge Detection



Original picture



Grayscale



Edge detection

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Practical work with Matlab

Image Processing Toolbox
Statistics and Machine Learning Toolbox

<https://www.mathworks.com/products.html>

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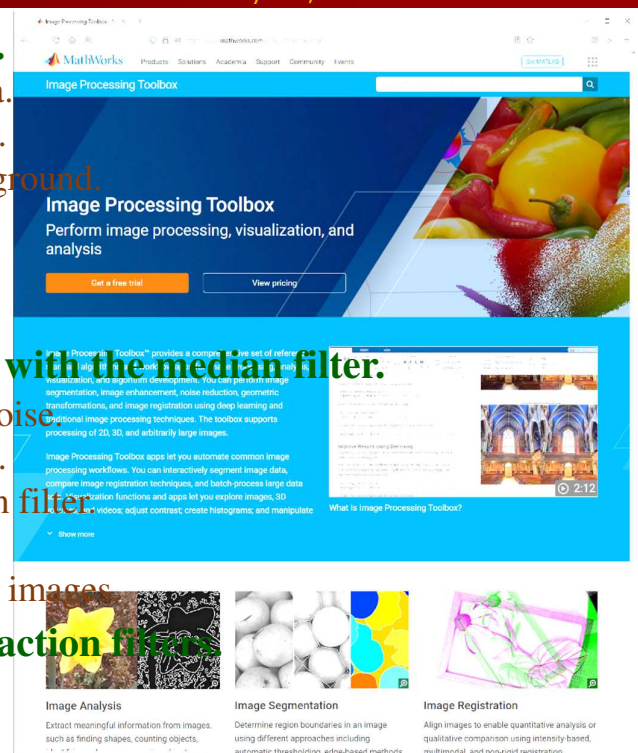
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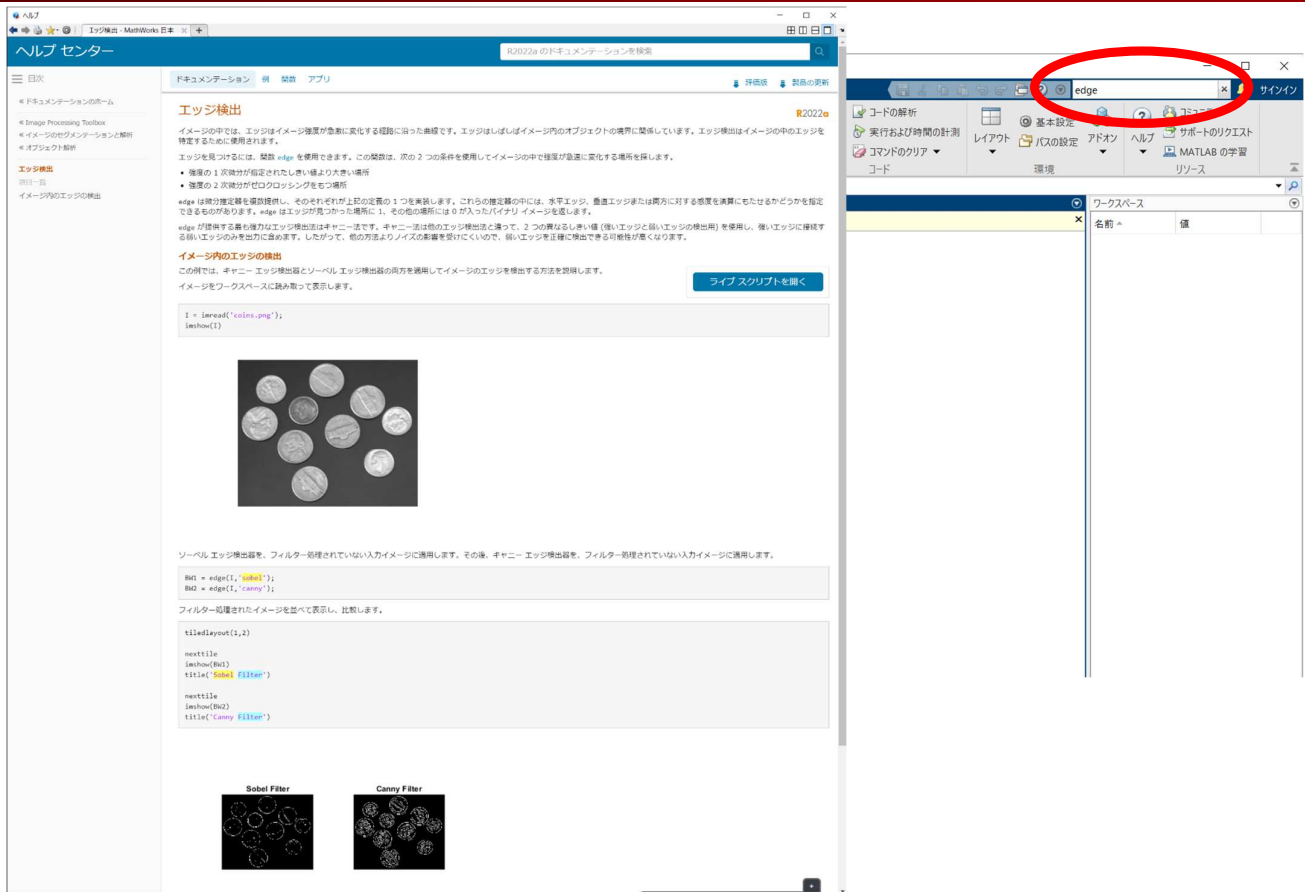
Practical work with Matlab

Image Processing Toolbox
Statistics and Machine Learning Toolbox



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<https://www.mathworks.com/products.html>
Physics, Informatics and Robotics



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Practical work with OpenCV-Python

https://docs.opencv.org/4.x/d6/d00/tutorial_py_root.html



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*Presented by
Hideyuki Sawada Laboratory
Waseda University*

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